

Bioinformatics for engineers

Alignment algorithms

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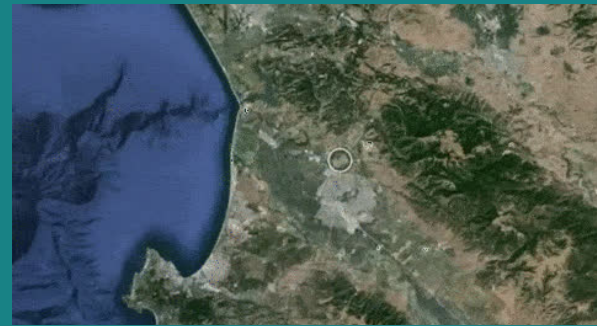
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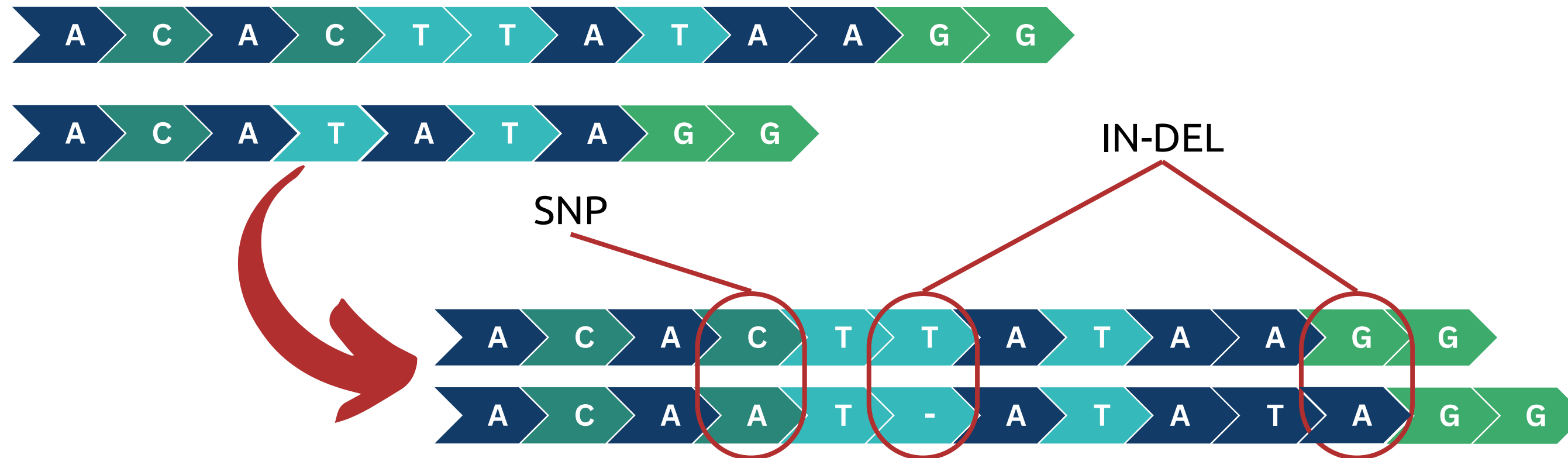
6. Divide and conquer
(BLAST)



WHAT IS ALIGNMENT?

The alignment

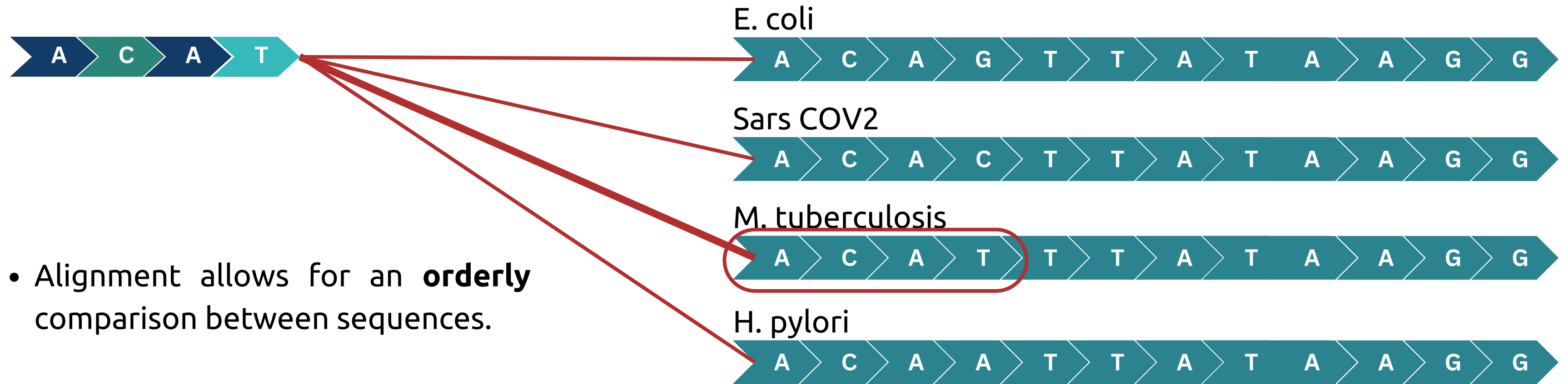
How to compare?



- Alignment allows us to identify congruent regions between misaligned biological sequences.
- Different mutations can lead to discrepancies in the genome that occur through insertions, deletions, and single nucleotide polymorphisms (SNPs).

SEQUENCE IDENTITY IDENTIFICATION

Why you should use sequence alignment?



- Alignment allows for an **orderly** comparison between sequences.
- It can be performed with both DNA and AA sequences, however the three-dimensional nature of proteins reduces its effectiveness in these cases.

GLOBAL TO LOCAL

The sequences have the same length?

Query



Global



assume that the lengths of the sequences are equal

Local



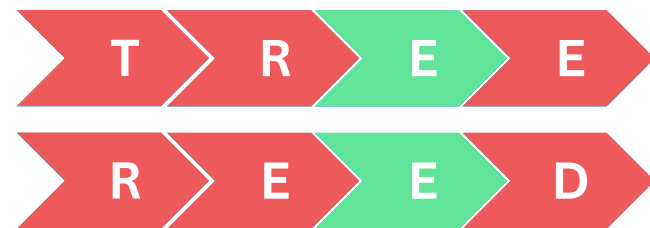
Assume that one of the sequences is a subunit of another

A quick example

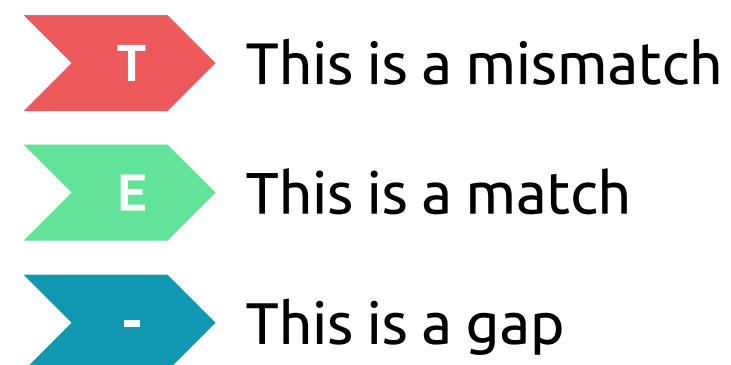
Two “sequences”



Two different alignments



Which one of these is the optimal?



Now you can assign a value to each one of these blocks

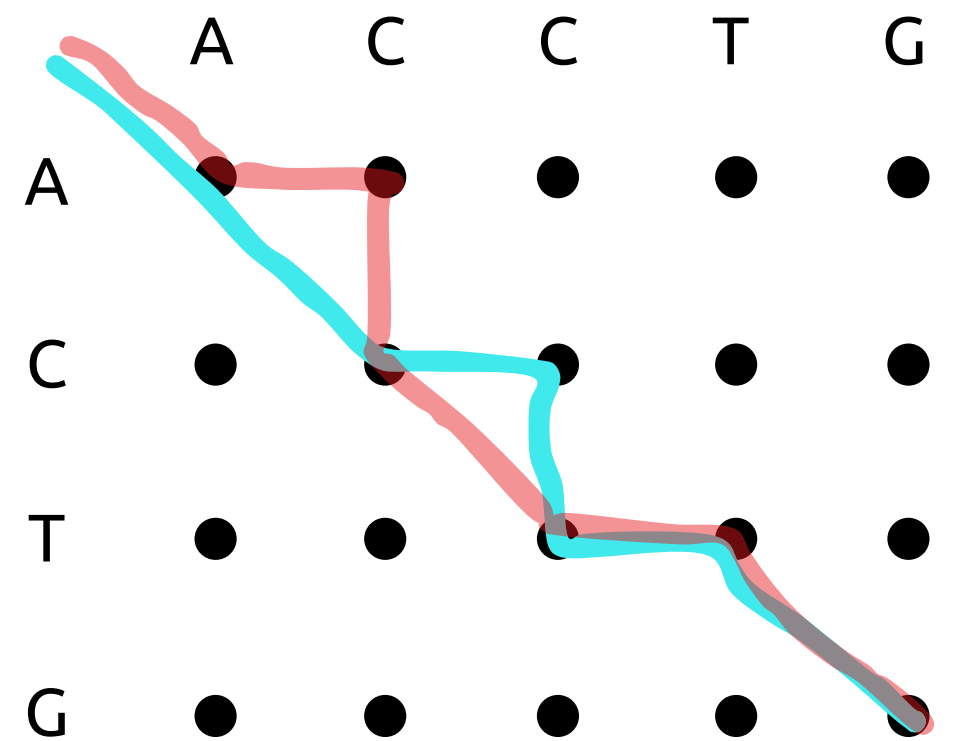
$$d(a,b) = \begin{cases} 1; a = b & \text{Match value} \\ \mu; a \neq b & \text{Mismatch penalty} \\ \delta; a = "-" ; b = "-" & \text{Gap penalty} \end{cases}$$

Through how we assign the mismatch and gap penalties, we can figure out what is the optimal alignment

The sequencing sudoku

Score matrix

This matrix lines up the two sequences to explore different alignment possibilities



For the blue path: 3 gaps, 3 matches

ACC-TG

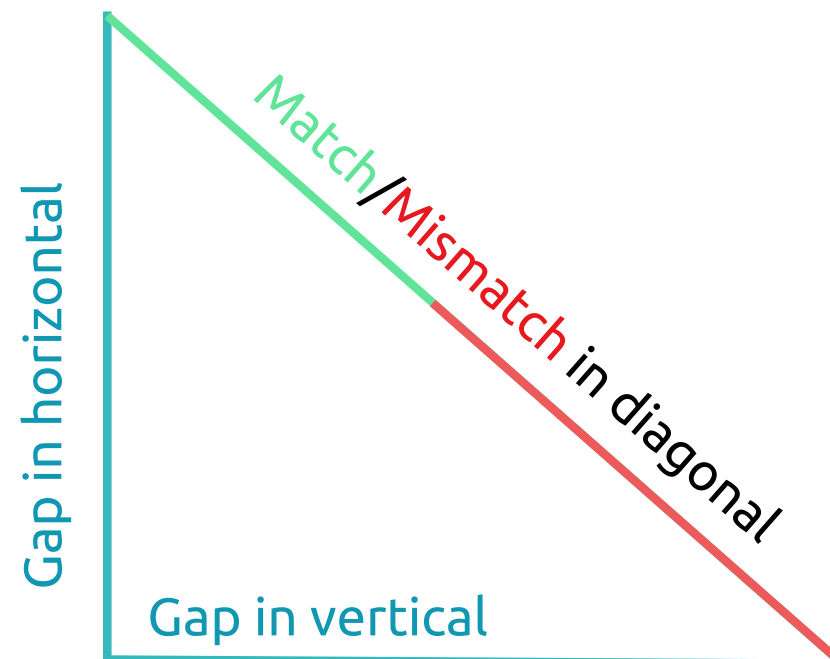
AC-T-G

For the red path: 3 gaps, 2 matches, 1 mismatch

AC-CTG

A-CT-G

So... between all these possibilities, which is the best one?



The sequencing sudoku

Score matrix

This is when scoring comes in handy...



		A	C	C	T
	0	-1	-2	-3	-4
A	-1	1			
C	-2				
T	-3				
G	-4				

$$\max(-1-1, -1-1, 0+1) = \max(-2, -2, 1) = 1$$

- Gap score = -1
- Mismatch score = -1
- Match score = 1

$$H(i, j) = \max \begin{cases} H(i-1, j-1) + S(a_i, b_j) \\ H(i-1, j) + d \\ H(i, j-1) + d \end{cases}$$

- Every alignment algorithm requires a priori a scoring matrix that indicates the compatibility of each nucleotide in sequence a with b.
- This process is iterative and can only be performed between two sequences.

Needlman Wunch algorithm

This algorithm seeks **global alignment**. It attempts to match each base in sequence A with each base in sequence B, from beginning to end.



The scoring matrix needs to be **initialized** with **accumulative gap** penalties in the first row and column.

	A	C	C	T	
A	0	-1	-2	-3	-4
C	-1	1	0	-1	-2
T	-2	0	2	1	0
G	-3	-1	1	1	2
	-4	-2	0	0	1

A	-	C	T	G
A	C	C	T	-

After you initialize your matrix, you are ready to **fill it up**, by evaluating each cell neighbor and choosing the path with the **maximum value**.

Finally, once the matrix is filled, you have to "traceback" your alignment. You ALWAYS start tracing back from the bottom right cell and moving up until you get to the (0,0) cell.

To know which one to choose, you have to "recalculate" which one, when you add its respective bonus or penalty, gives you the value of your current cell.

In this example, the gap penalty is -1, as well as the mismatch penalty. How much would it change with these new values: Gap (-4), Mismatch (-3), Match (1)

Smith Waterman algorithm

This algorithm seeks **local alignment**. It doesn't care about the beginning or the end; he just looks for the piece of sequence where the resemblance is greatest.

The scoring matrix needs to be **initialized** with **zeros** penalties in the first row and column.

	A	C	C	T
A	0	1	2	3
C	0	2	3	4
T	0	3	4	5
G	0	4	5	6

After you initialize your matrix, you are ready to **fill it up**, by evaluating each cell neighbor and choosing the path with the **maximum value BUT now, if you get a negative value you replace it with a zero.**

Positive
Vibes
Only

Once the matrix is filled, you will traceback starting from the cell with the highest score in the entire matrix and stop when you reach 0.

See how the alignment changes using this algorithm.

A	C	T	G	-	-	-
A	-	-	-	C	C	T

DIVIDE AND CONQUER

What is BLAST?

Basic Local Alignment Search Tool

If the Smith Waterman algorithm is so exact, ¿why does BLAST exists?

Algorithmic complexity $O(n \times m)$

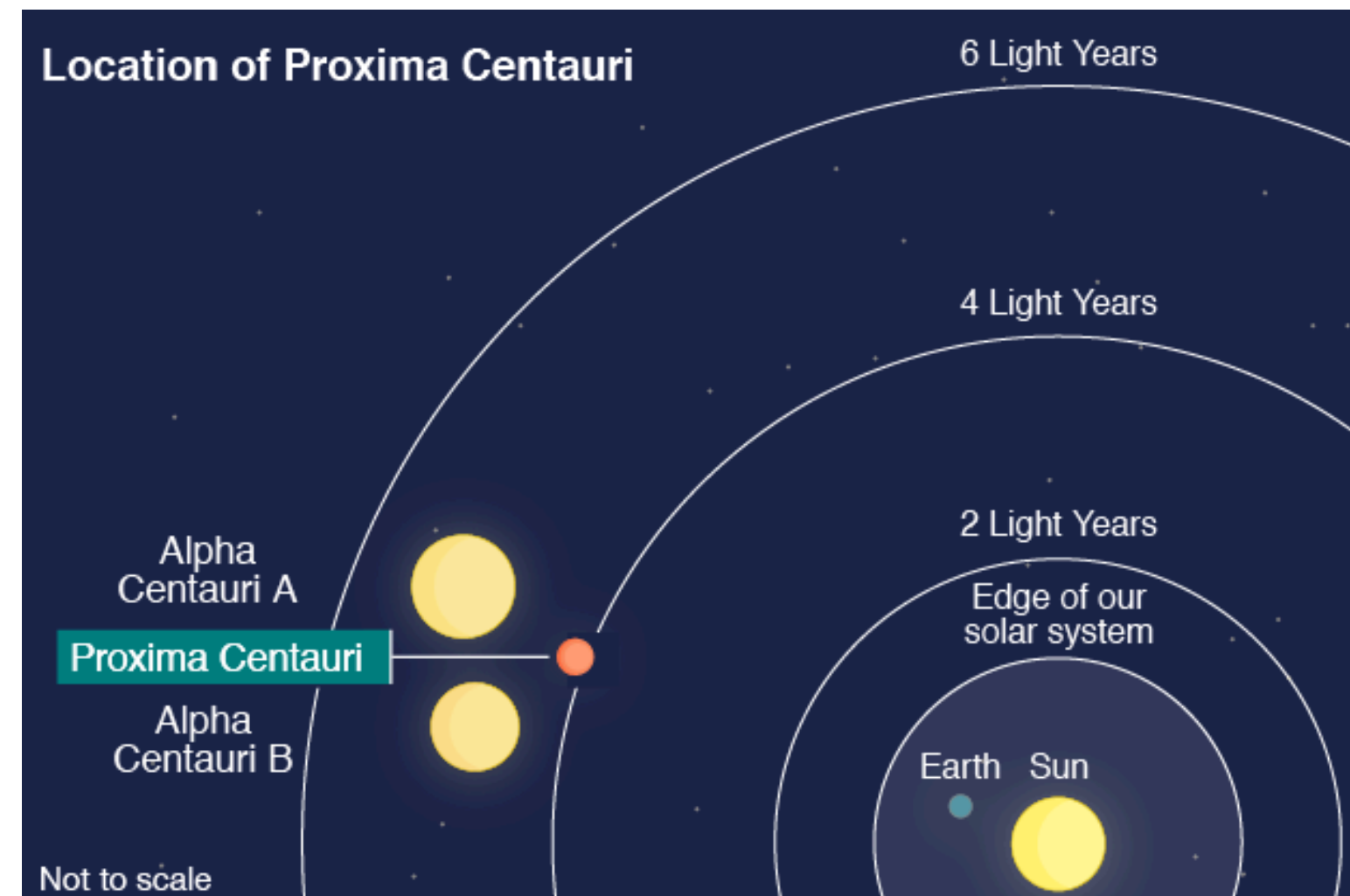
Imagine you have a **query sequence of length n** and you want to search for matches in a **database of total length m** .

For each cell in the matrix, the computer must perform **at least 3 comparisons and one sum** (look up, left, diagonal and sum the score).

Assume a length n of 10000pb and a length m GenBank database with a subset of 10^{11} pb.

Total operations = $10^3 * 10^{11} = 10^{14}$ seconds

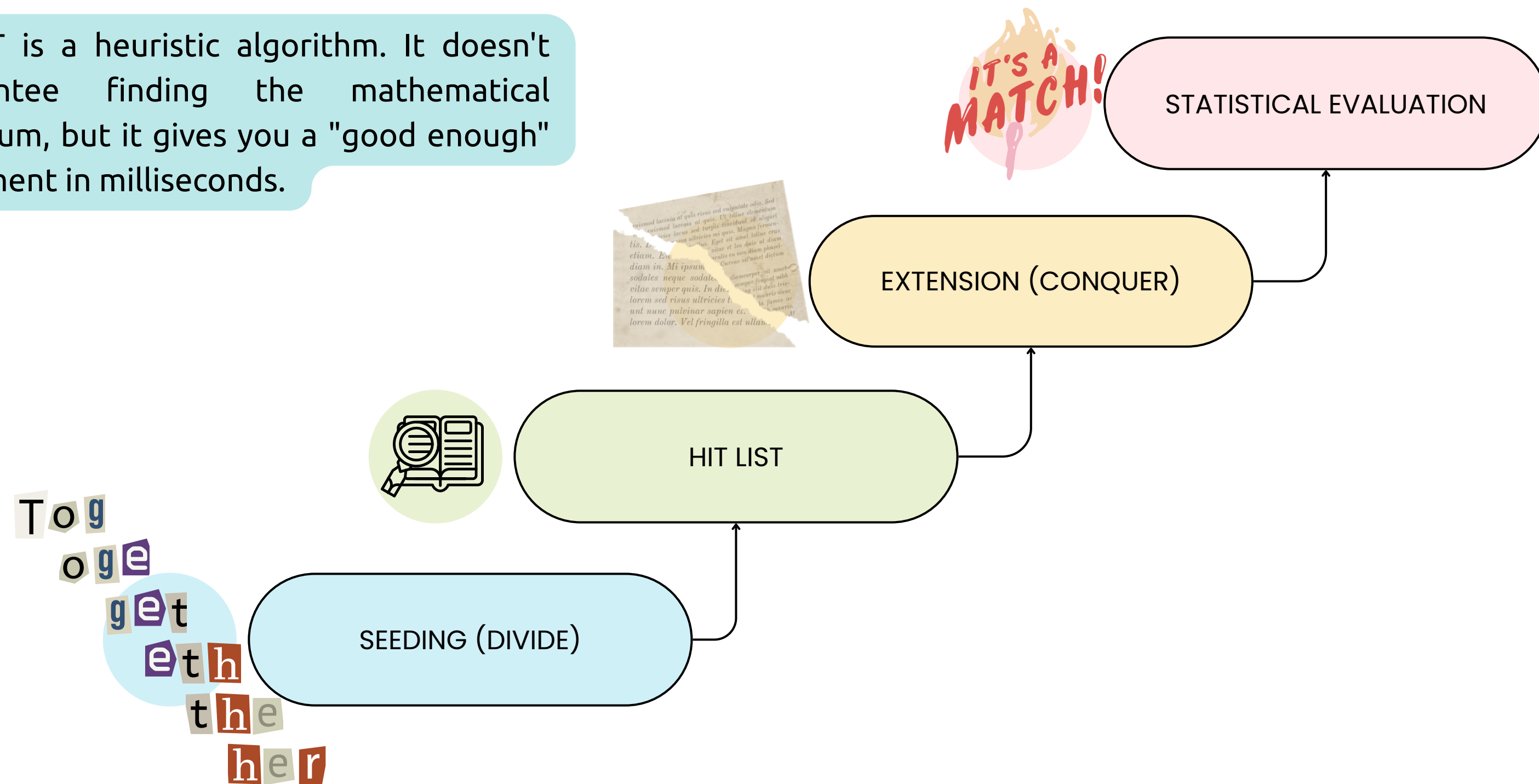
3,170,979 years, aproximately :)



As you can see, if you try to make this happen with software, you'd reach proxima centauri, travelling at 17 km/s, before the alignment ends, with 3M years to spare.

What does Napoleon have to do with this?

BLAST is a heuristic algorithm. It doesn't guarantee finding the mathematical optimum, but it gives you a "good enough" alignment in milliseconds.



Dividing and conquering: Seeds

AACCGGTTA

AACC

ACCG

CCGG

CGGT

GGTT

GTTA

BLAST takes k-mers, which are just small pieces out of our sequence of equal k length as *seeds*.

This way, it is way easier to look for our sequence in a database.

Imagine this is your database, and you will be using, out of many, this 4-mer

GGTT

BLAST will look first, for *hits* of only this particular seed.

TGCAAACCGTAGTTAAAGCTGAAC**GGTT**AGCGTTGCA**GGTT**AAAC
CGTAGC**GGTT**CAACCGGTTAACGTTGCAGGTTGCCGGTTAACGT**G**
GTTAAGCTGA AACCCCAACC**GGTT**AAGCTGACCGTACGTCCTAA
TTTACGGCATGCATGGGTACCAGTCCGTGGTCAAATGCACGTCGTC
ACTGCTAGCTGGCATGCTAGCTGCGTGTATTCGGATT**GGTT**GTACC

Dividing and conquering: extension

AACCGGTTA

AACC

ACCG

CCGG

CGGT

GGTT

GTTA

Then, will make it *grow, expanding it* left and right.

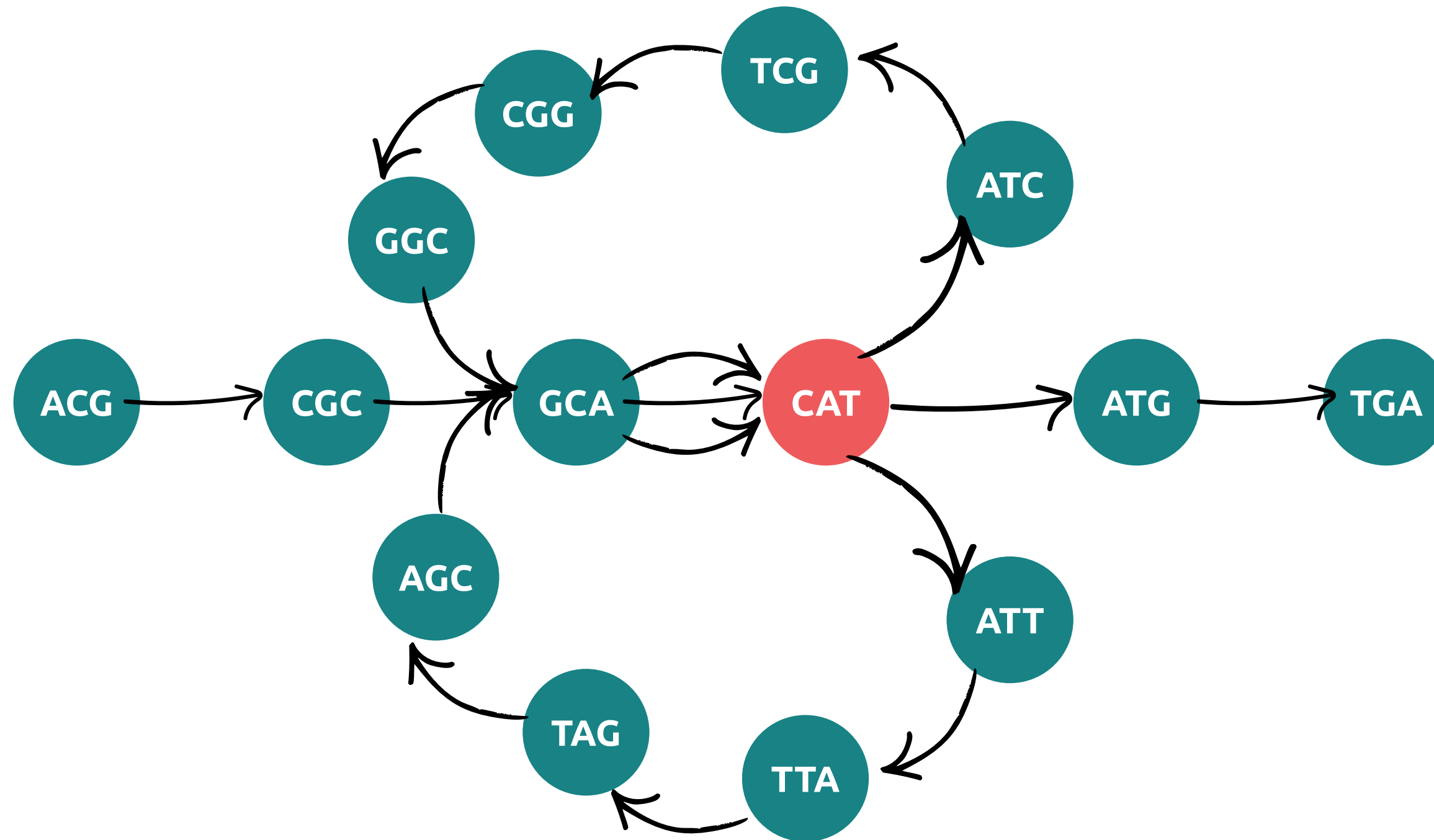
After extending it, it evaluates if there is a MATCH or MISMATCH between the sequence and the database

TGCAAACCGTAGTTAAAGCTGAACGGTTAAGCGTTGCAGGTTAAACCG
TAGCGGTTCAACCGGTTAACGTTGCAGGTTGCCGGTTAACGTGGTTA
AGCTGA AACCCCAACCGGTTAAGCTGACCGTACGTCCTAA
TTACGGCATGCATGGGTACCAGTCCGTGGTCAAATGCACGTCGTC
ACTGCTAGCTGGCATGCTAGCTGCGTGTATTCGGATTGGTTGTACC

You can see that only one region has a 100% match with our sequence, so now you know where this particular sequence is more likely to belong.

Dividing: De Bruijn graphs

Visualize what we get from k-mers, pay close attention, we'll use this later



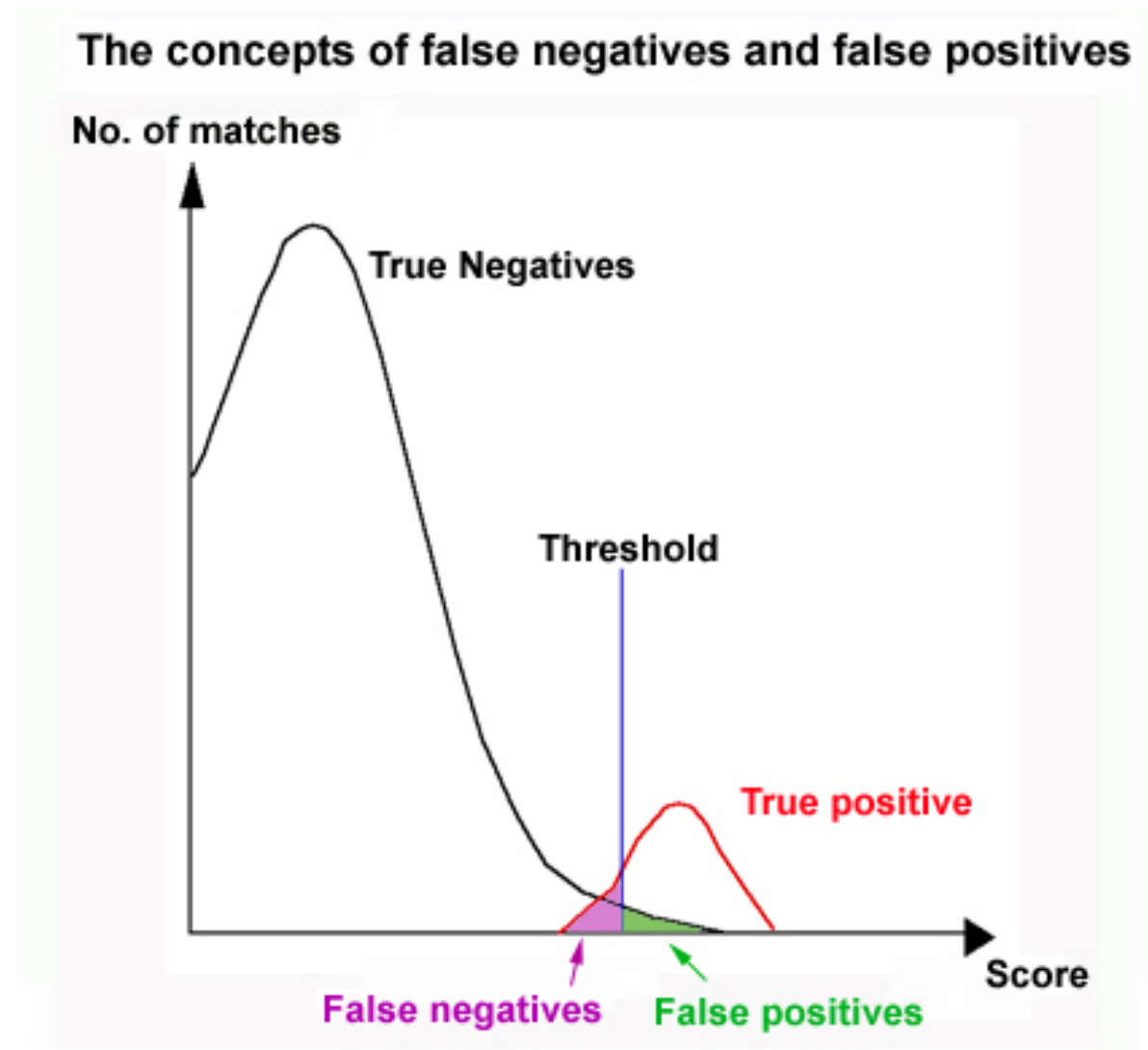
How do we know if a match is real?

Not all matches are real; **some are pure luck**. BLAST calculates the E-value (Expectation Value) for evaluating how real a match is.

The E-value is the expectation value that indicates the number of alignments with a score $\geq S$ that one can expect to find by chance in a database of size N .

In simpler words, if you expect to find this match almost EVERYWHERE, then most probably your match is just pure luck

We want to find low e-values for our alignments, this means our results are not coincidences.



The BLAST flexibility

You can adjust your query and database depending on your needs, thus there are different types of BLAST.

Name	Query	Database	Alignment	Utility
BLASTn	Nucleotide	Nucleotide	Nucleotide	Finding identical genes, mapping genomes, or searching for contaminants.
BLASTp	Aminoacids	Proteins	Protein	Sequence identification and similarity searches
BLASTx	Translated nucleotides	Proteins	Protein	Identification of potential proteins encoded by query sequences; detection of novel viruses
tBLASTn	Proteins	Nucleotide (translated to protein)	Protein	Identification of database sequences encoding proteins similar to the query
tBLASTx	Translated nucleotides	Nucleotide (translated to protein)	Protein	Identification of nucleotide sequences similar to the query based on their coding potential

From: <https://chanzuckerberg.zendesk.com/hc/en-us/articles/360050963352-A-Guide-to-BLAST>

Summing up

What is the main difference between BLAST and alignment algorithms like Smith Waterman?



Smith Waterman will have to look for Waldo checking each person in that city.

BLAST will just call anyone wearing a red-striped shirt, check their IDs and find Waldo.



That will definitely take less time...

Thanks !!!!!